Chapter 3
Data-link layer

Functions of data-link layer:-
The **Data Link Layer** is the **second layer of the OSI model**. Its main function is to make sure that data travels safely and correctly between two directly connected devices on the same network.
The following functions of data-link layer are:
Framing: When we send a file, the Data Link Layer breaks the data into frames before sending it over the network.

Physical addressing: Each device on a network has a unique **MAC Address** (Physical Address). The Data Link Layer adds the sender's and receiver's MAC addresses to each frame.
Example:
When our computer sends data to a printer on the same network, the printer's MAC address is used to deliver the frame to the correct device.

Flow control: If the rate at which the data are consumed by the receiver is less than the rate produced by the sender, the data link layer deals with a flow control mechanism to prevent overrun the receiver.

Error control: Sometimes data can be damaged during transmission because of noise or interference. The Data Link Layer checks whether the received frame contains errors. If a frame is corrupted during transmission, the receiver can request that it be sent again.

Access control: When many devices share the same communication channel, the Data Link Layer decides which device can send data at a particular time.
Example:
In a Wi-Fi network, many devices are connected to the same wireless medium. Access control helps avoid transmission conflicts.

Reliable Delivery: The Data Link Layer tries to ensure that frames reach the destination correctly. It uses acknowledgments and retransmission when necessary.

Overview of logical link control (LLC) and Media access control (MAC)
Logical Link Control (LLC):
LLC is a sub-layer of the Data Link Layer that provides services to the Network Layer and controls how data is sent over the network.
Functions:
1. Protocol Identification

- LLC identifies which network protocol is using the data.

- It separates different types of data like IP, IPX, etc.

2. Flow Control

- LLC controls the speed of data transfer between sender and receiver.
- It prevents fast sender from overwhelming slow receiver.

3. Error Control

- LLC checks whether data is received correctly.
- If data is damaged, it may request retransmission.

4. Interface Function

- LLC acts as a bridge between:
- Network Layer (software side)
- MAC Layer (hardware side)

Protocols: Some common LLC protocols include:

- High-Level Data Link Control (HDLC)
- Point-to-Point Protocol (PPP)
- Link Access Procedure, Balanced (LAPB)

Media Access Control (MAC):
A MAC address is a unique hardware address assigned to a device that is used to identify. It is used to identify devices inside a local network (LAN or Wi-Fi).
It helps to identify exactly which device is sending or receiving data. It is also known as physical layer or hardware layer. A MAC address is a **48-bit address** written in hexadecimal form. It is Stored in the Network Interface Card (NIC)

Functions:
The MAC sub-layer is the lower part of the Data Link Layer. Its main task is to control how devices share and use the network.
1. Physical Addressing

- MAC uses **MAC addresses** to identify sender and receiver devices.
- It ensures data goes to the correct device.

2. Channel Access Control

- It avoids multiple devices sending data at the same time.

3. Framing (Frame Formation)
MAC prepares data into frames before sending it.

It adds:
- Header (address info)
- Trailer (error checking)

4. Error Detection

- MAC adds error-checking code (like CRC) to detect damaged data.
- If data is corrupted, it is discarded.

5. Collision Control

- Only one device sends data at a time
- Or it detects collision and fixes it

Protocols: Some common MAC protocols include:

- Ethernet (IEEE 802.3)
- Wi-Fi (IEEE 802.11)
- Token Ring (IEEE 802.5)

Framing:
Framing is the process of dividing a data stream into smaller, manageable units called frames. Each frame is a self-contained packet of data that includes:

1. **Header**: Contains control information such as source and destination addresses, sequence numbers, and error-checking data.
2. **Data**: The actual payload of the frame.
3. **Trailer**: Typically includes error-checking data, such as a checksum or CRC (Cyclic Redundancy Check).

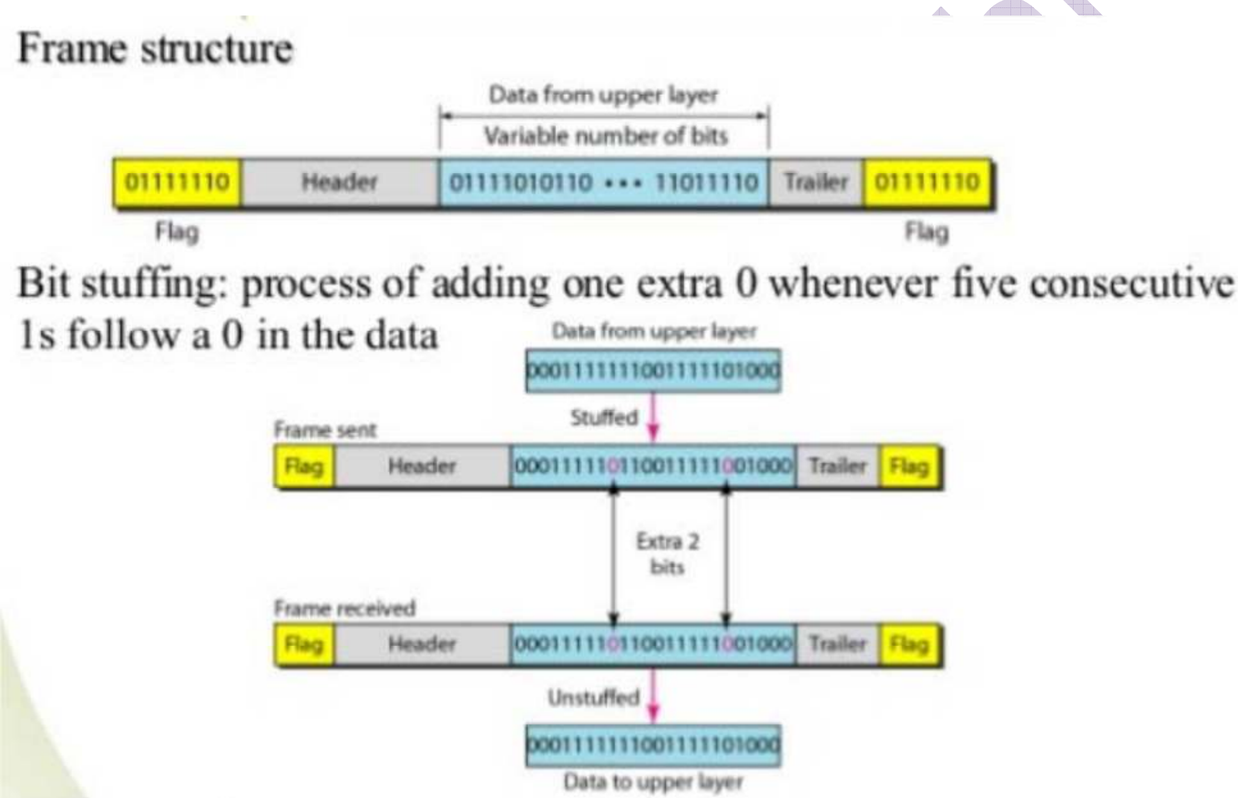
Framing serves several purposes:

- **Error detection and correction**: By including error-checking data, frames can be checked for errors during transmission.
- **Flow control**: Framing helps regulate the amount of data that can be sent at one time, preventing network congestion.

- **Multiplexing**: Multiple frames can be transmitted over the same channel, allowing multiple devices to share the same communication medium.

Header	Payload	Trailer	Header	Payload	Trailer
Frame 1			Frame 2		

Figure : Frame Format in Variable Size frame



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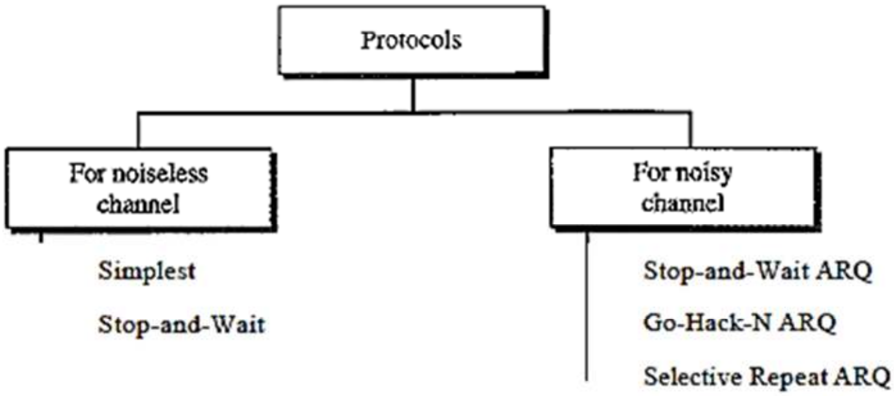
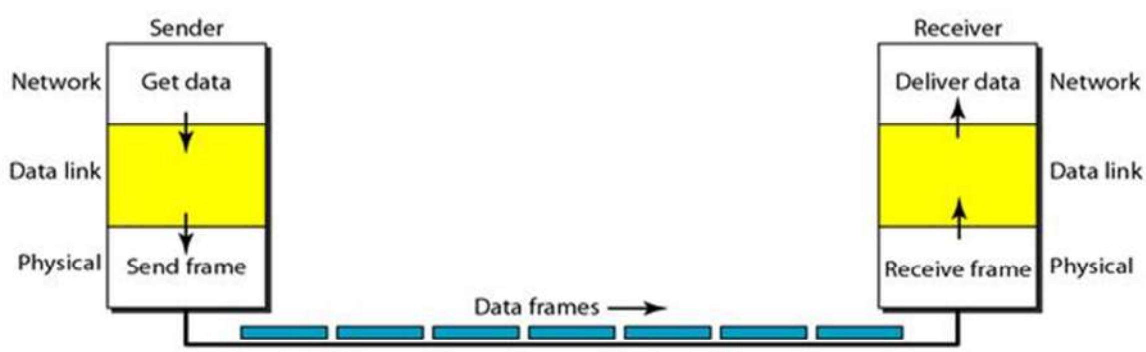
Flow control:- Flow control mechanisms regulate the rate at which data is sent over a network to prevent congestion, buffer overflows, and data loss.



For noiseless channel:-
Simple:- Simplest Protocol is a unidirectional protocol in which data frames are traveling in only one direction-from the sender to receiver.
The sender sends a sequence of frames without even thinking about the receiver. Data are transmitted in one direction only. Both sender & receiver always ready. Processing time can be ignored. Infinite buffer space is available. And best of all, the communication channel between the data link layers never damages or loses frames. This thoroughly unrealistic protocol.
It does not handle either flow control or error correction.

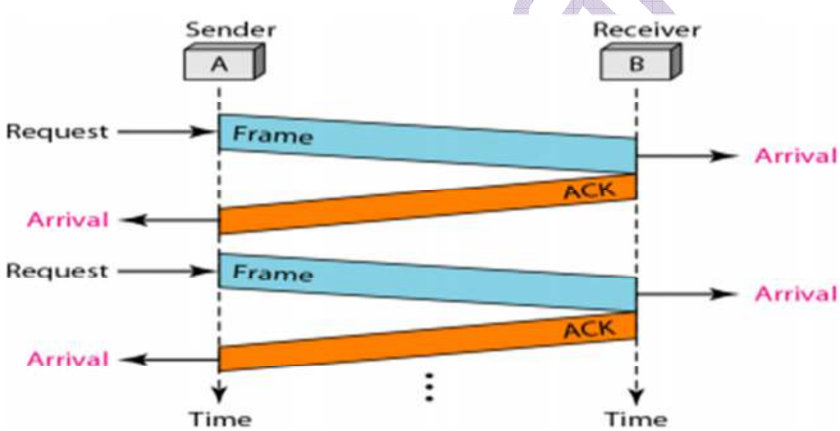


Stop-and-wait Protocol:- the sender sends one frame, stops until it receives confirmation from the receiver (okay to go ahead), and then sends the next frame. It has unidirectional communication for data frame.
Advantage:
The main advantage of stop & wait protocol is its accuracy.

- It sends the next data packet only when the acknowledgment of the previous one has been received so that there is no possibility to loss data.

Disadvantage:

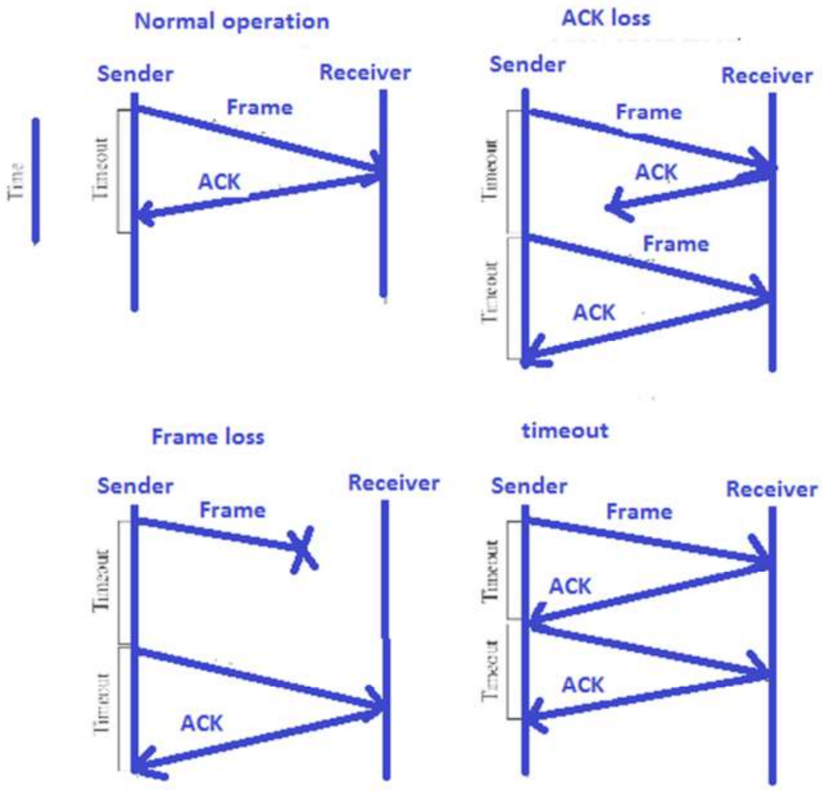
- The sender and receiver wait infinite time to send data and receive acknowledgement. if sender send the packet and receiver not reply acknowledgement then there arises deadlock .
  1. Problems occur due to lost data
     - Sender waits for an infinite amount of time for an acknowledgment.
     - Receiver waits for an infinite amount of time for a data.
  2. Problems occur due to lost acknowledgment
     - Sender waits for an infinite amount of time for an acknowledgment.
  3. Problem due to the delayed data or acknowledgment
     - the acknowledgment is received after the timeout period on the sender's side.



For Noisy channal:

1 Stop-and-Wait Automatic Repeat Request:-

Stop-and-wait ARQ is based on the stop-and-wait flow control technique

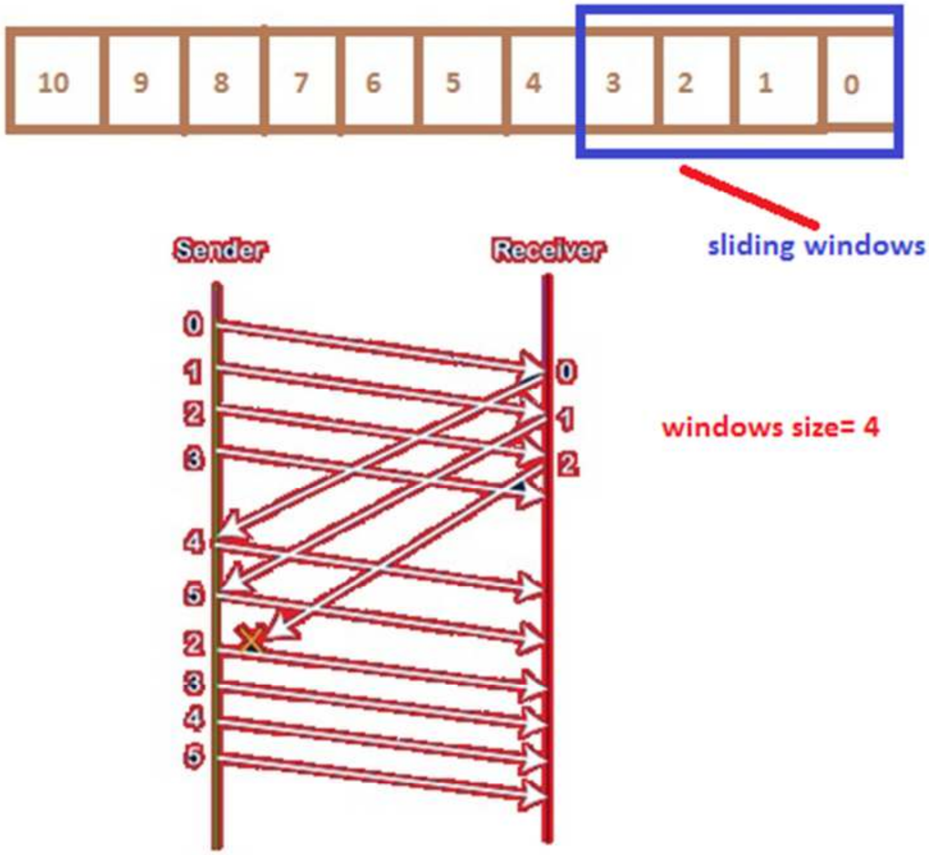


1. The sender maintains a timeout counter.
2. When a frame is sent, the sender starts the timeout counter.
3. If acknowledgement of frame comes in time, the sender transmits the next frame in queue.
4. If acknowledgement does not come in time, the sender assumes that either the frame or its acknowledgement is lost in transit. Sender retransmits the frame and starts the timeout counter.
5. If a negative acknowledgement is received, the sender retransmits the frame.

Go-Back-N ARQ:

Important points related to Go-Back-N ARQ:

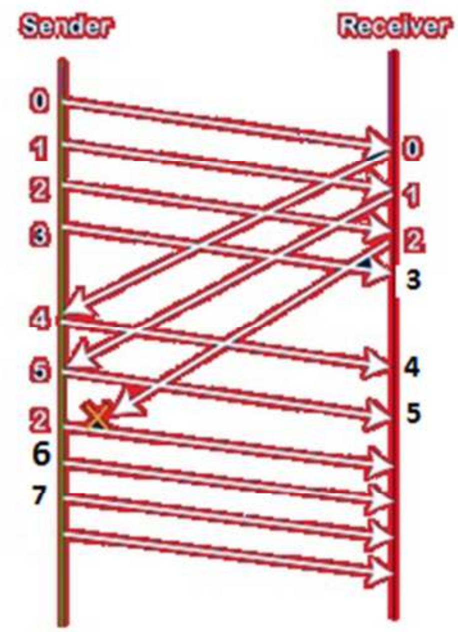
- In Go-Back-N, N determines the sender's window size, and the size of the receiver's window is always 1.
- It does not consider the corrupted frames and simply discards them.
- It does not accept the frames which are out of order and discards them.
- If the sender does not receive the acknowledgment, it leads to the retransmission of all the current window frames.



Selective Reprat ARQ :-

Selective repeat is also the sliding window protocol which detects or corrects the error occurred in data link layer. The selective repeat protocol retransmits only that frame which is damaged or lost. In selective repeat protocol, the retransmitted framed is received out of sequence. The selective repeat protocol can perform following actions.

- The receiver is capable of sorting the frame in a proper sequence, as it receives there transmitted frame whose sequence is out of order of the receiving frame.
- The sender must be capable of searching the frame for which the NAK has been received.
- The receiver must contain the buffer to store all the previously received frame on hold till the retransmitted frame is sorted and placed in a proper sequence.
- The ACK number, like NAK number, refers to the frame which is lost or damaged.
- It requires the less window size as compared to go-back-n protocol.

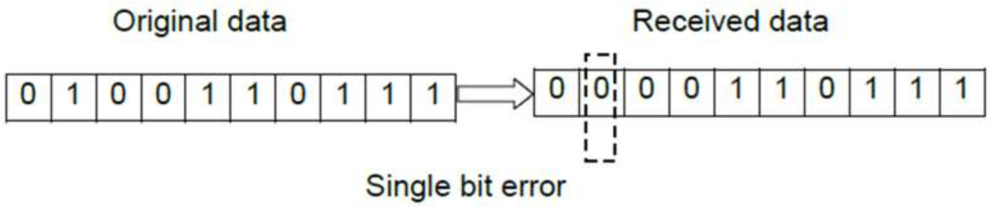


Error control:- Error control is the process of detecting and correcting both the bit level and packet level errors or Error control is both error detection and error correction. It allows the receiver to inform the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender. In the data link layer, the term error control refers primarily to methods of error detection and retransmission. Error control in the data link layer is often implemented simply:

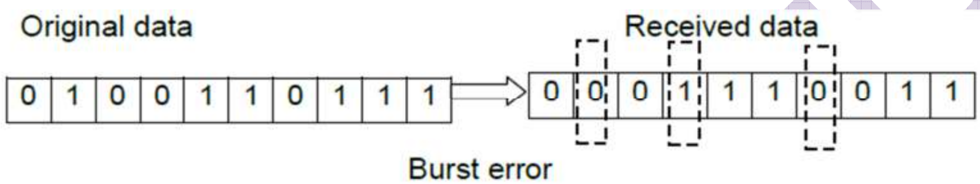
Types of error:- Whenever bits flow from one point to another, they are subject to unpredictable changes because of interference. This interference can change the shape of the signal. There are two types of error.

1. Single bit errors
2. Burst Errors

1.Single bit errors :- In single-bit error, a bit value of 0 changes to bit value 1 or vice versa. Single bit errors are more likely to occur in parallel transmission.



2.Burst errors:- The term burst error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1.



Error Detection Techniques:-There are different techniques used for error detection and error correction. The most popular Error Detecting Techniques are:

- Single parity check
- Two-dimensional parity check
- Checksum
- Cyclic redundancy check

Single parity check:-

- Single Parity checking is the simple mechanism and inexpensive to detect the errors.
- In this technique, a redundant bit is also known as a parity bit which is appended at the end of the data unit so that the number of 1's becomes even.
- If the number of 1s bits is odd, then parity bit 1 is appended and if the number of 1s bits is even, then parity bit 0 is appended at the end of the data unit.
- At the receiving end, the parity bit is calculated from the received data bits and compared with the received parity bit.

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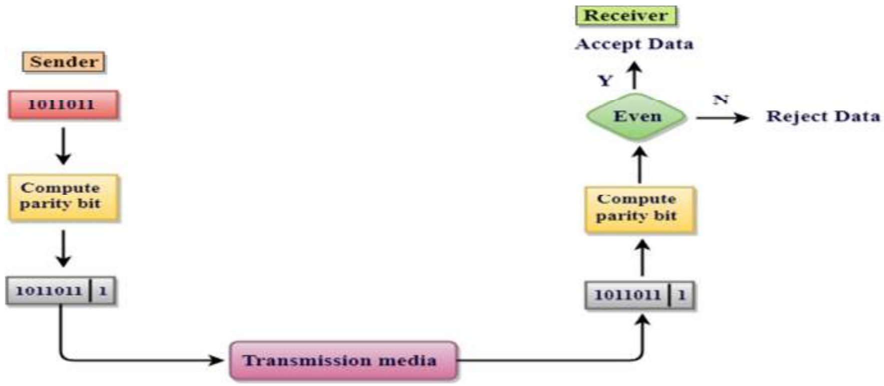
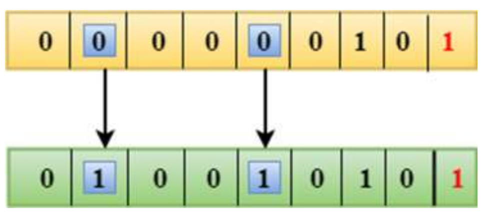
- This technique generates the total number of 1s even, so it is known as even-parity checking.



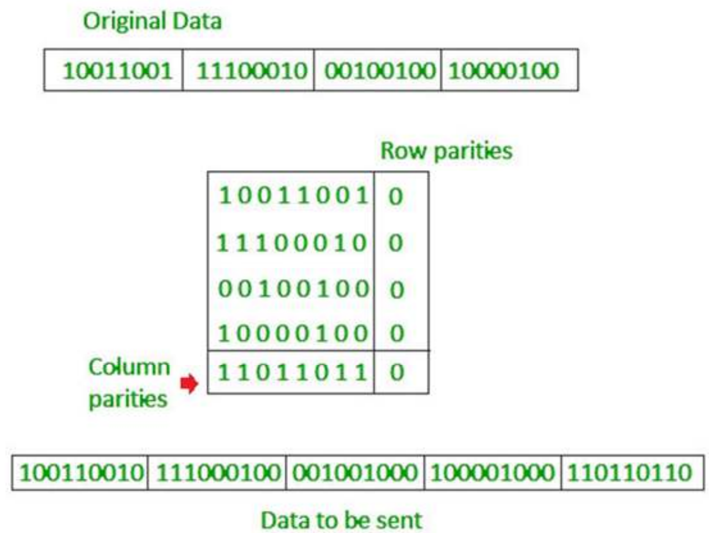
Drawbacks Of Single Parity Checking:-

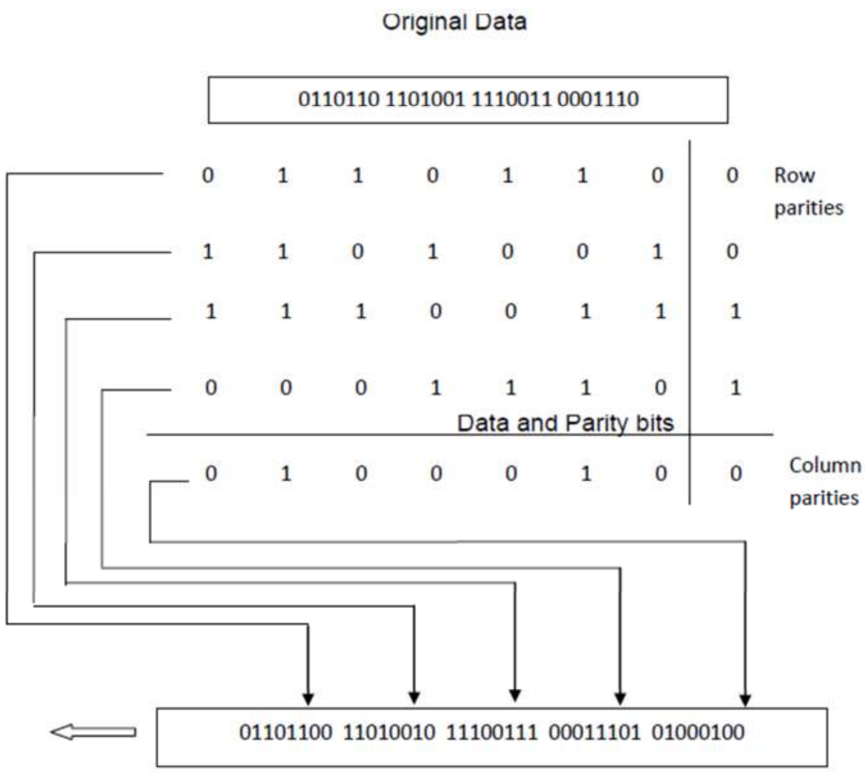
- It can only detect single-bit errors which are very rare.
- If two bits are interchanged, then it cannot detect the errors.



Two-dimensional Parity check:-

Parity check bits are calculated for each row, which is equivalent to a simple parity check bit. Parity check bits are also calculated for all columns, then both are sent along with the data. At the receiving end these are compared with the parity bits calculated on the received data.





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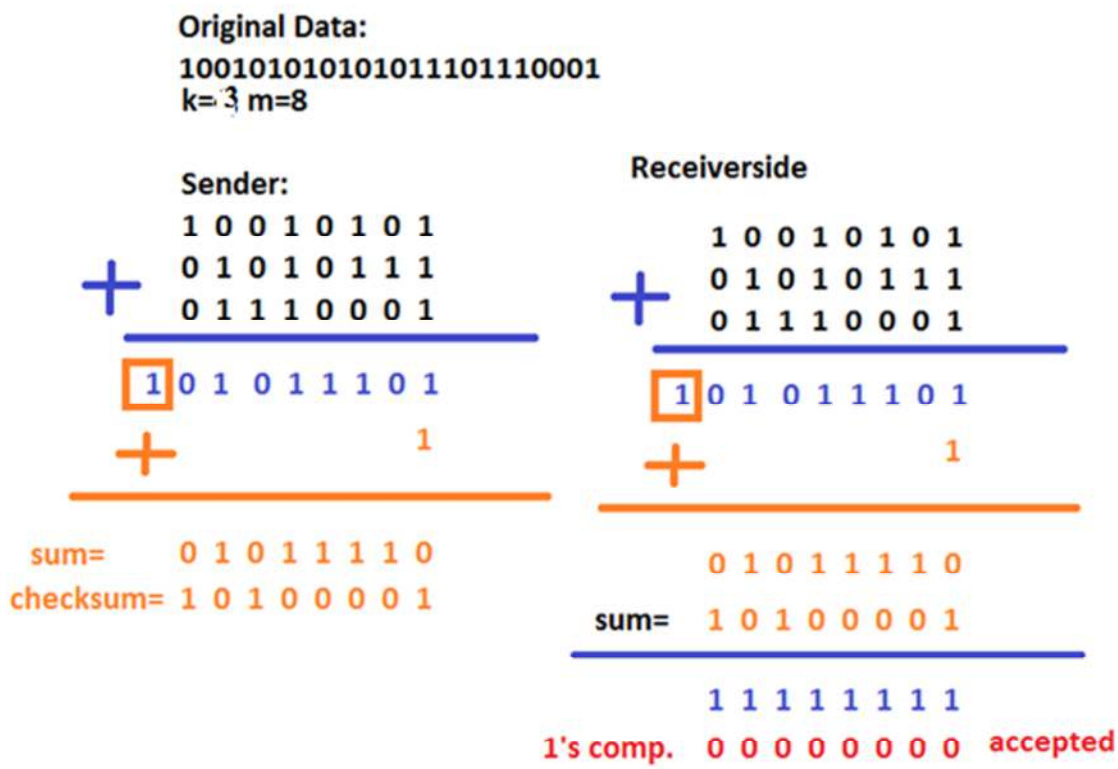
Checksum:-

- In checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.



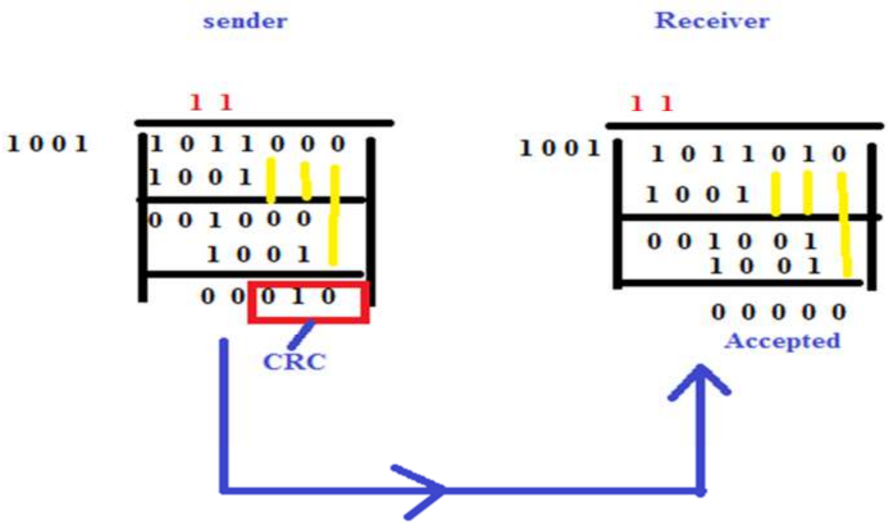
Cyclic redundancy check (CRC):- Most powerful of the redundancy checking techniques is the cyclic redundancy check (CRC). This method is based on the binary division

- In CRC technique, a string of n 0s is appended to the data unit, and this n number is less than the number of bits in a predetermined number, known as division which is n+1 bits.
- Secondly, the newly extended data is divided by a divisor using a process is known as binary division. The remainder generated from this division is known as CRC remainder.
- Thirdly, the CRC remainder replaces the appended 0s at the end of the original data. This newly generated unit is sent to the receiver.

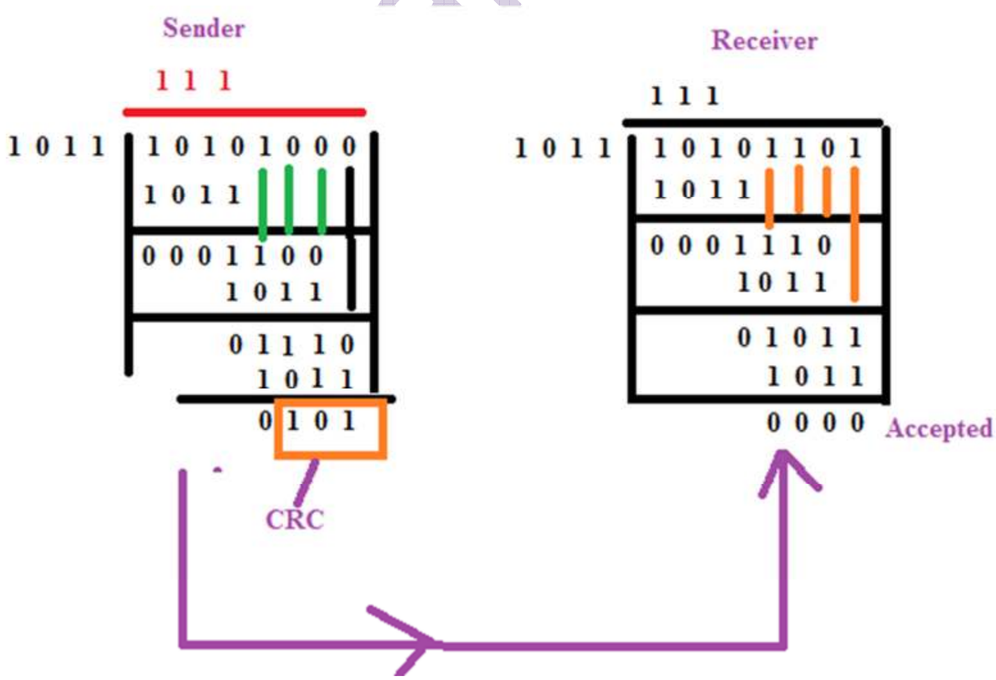
- The receiver receives the data followed by the CRC remainder. The receiver will treat this whole unit as a single unit, and it is divided by the same divisor that was used to find the CRC remainder.
- If the resultant of this division is zero which means that it has no error, and the data is accepted.

Q. if polinomial equation is x^3+x+1 and divisor is x^3+1.

1. first find equation's coeefficient $x^3+x+1= 1.x^3+0.x^2+1.x+1= 1011$
2. find the coefficient's of divisor $x^3+1= 1.x^3+0.x^2+0.x+1= 1001$



Data unit 1 0 1 0 1 is divided by 1011.



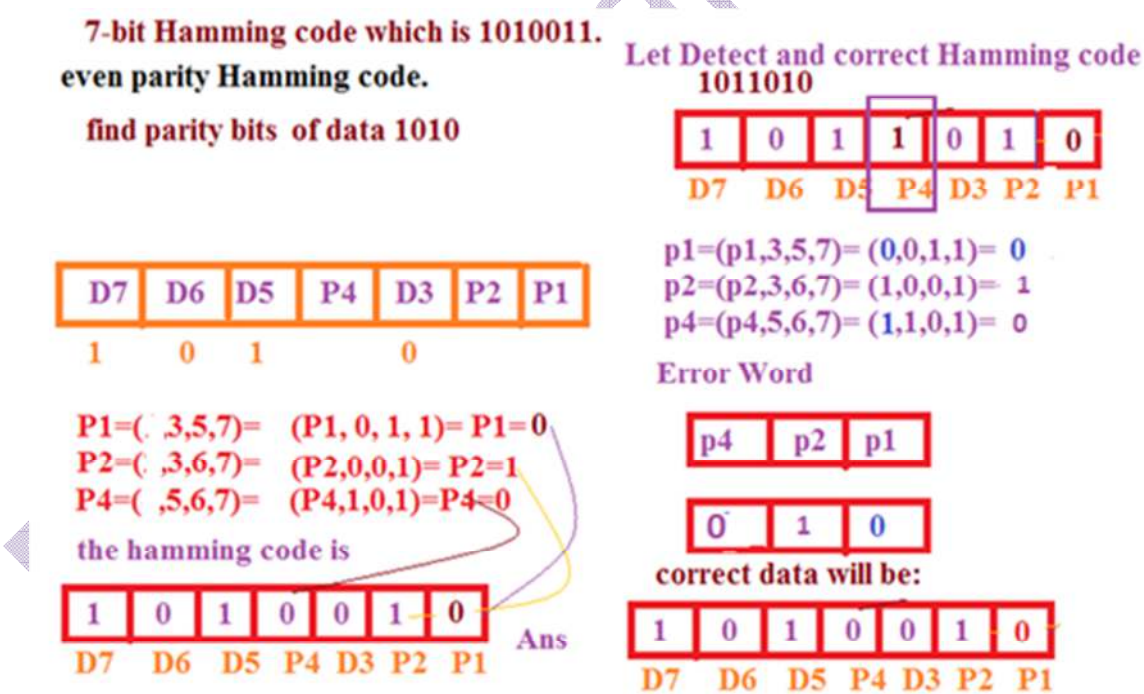
Error correction technique:- Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is moved or stored from the sender to the receiver. It is technique developed by R.W. Hamming for error correction.

the first step is to identify the bit position of the data & all the bit positions which are powers of 2 are marked as parity bits (e.g. 1, 2, 4, 8, etc.). The following image will help in visualizing the received hamming code of 7 bits.

Rules:-

1. Find the p1,p2 and p4 by the formula 2^n where n= 0,1,2,3,4…….
2. There are 7 columns in a table for store data and parity.
3. The values of p1, p2, and p3 is

P1=(p1,d3,d5,d6)=(1,3,5,7)
P2=(p2,d3,d6,d7)=(2,3,6,7)
P4=(p4,d5,d6,d7)=(4,5.6.7)



Channel Allocation Problem:-

Channel allocation is a process in which a single channel is divided and allotted to multiple users in order to carry user specific tasks. There are user's quantity may vary every time the process takes place. Channel allocation problem can be solved by two schemes: Static Channel Allocation in LANs and MANs, and Dynamic Channel Allocation.

1.Static Channel Allocation
2.Dynamic Channel Allocation

1.Static Channel Allocation:- It is the classical or traditional approach of allocating a single channel among multiple competing users Frequency Division Multiplexing (FDM). if there are N users, the bandwidth is divided into N equal sized portions each user being assigned one portion. since each user has a private frequency band, there is no interface between users.
It is not efficient to divide into fixed number of chunks.
In static channel allocation, the network bandwidth is divided equally between devices and is permanent for devices.

- For example, there are 50 users on the network, and the network bandwidth is 100 Hz. The 100 Hz bandwidth would be divided into 50 equally sized parts that are 2 Hz. Each user will get a 2 Hz portion.
- Each device has a private frequency band, so there is no possibility of interference with other devices.
- A well-known example of a single channel allocation scheme is FM radio, in which different frequencies are assigned to different stations that are fixed.

T = 1/(U*C-L)

T(FDM) = N*T(1/U(C/N)-L/N)
Where,

T = mean time delay,
C = capacity of channel,
L = arrival rate of frames,
1/U = bits/frame,
N = number of sub channels,
T(FDM) = Frequency Division Multiplexing Time

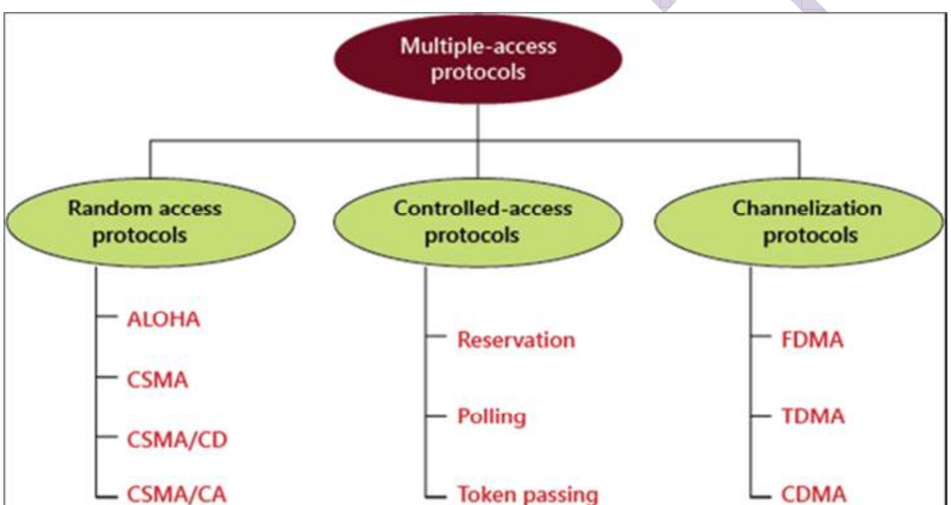
2.Dynamic Channel Allocation:-

Possible assumptions include:

To overcome this problem, we can use dynamic channel allocation, in which the bandwidth is not allocated to the device permanently.

- In dynamic channel allocation, the bandwidth is not allocated to the device permanently, the frequency band is allocated to the device whenever required.
- When the device completes its communication on a channel, it is de-allocated, and the same channel can be assigned to another device.
- In dynamic channel allocation, a channel is dynamically allocated between devices.

Media Access(Multiple Access):-
The media access control protocol is divided into three protocols: The Random access, the Controlled-access, and the Channelization protocol.



Random Access Protocol:-
In the Random access protocol, all systems are equal. No anyone system can depend and control another system. However, if more than one station attempts to transmit the data, there is an access conflict—collision, due to which the frames are either lost or changed.

Random access protocol is divided into two categories; firstly, aloha, and second CSMA (carrier sense multiple access). CSMA is later further divided into two parallel methods; CSMA/CD and CSMA/CA. When a collision is detected, CSMA/CD tells the station what to do. CSMA/CA attempts to stop a collision.

Aloha:-
Aloha protocol was developed by Abramson at University of Hawaii. In the Hawaiian language, Aloha means affection, peace, and compassion. University of Hawaii consists of a number of islands and obviously they cannot setup wired network in these islands. In the University of Hawaii, there was a centralized computer and there were terminals distributed to different islands. It was necessary for the central computer to communicate with the terminals and for that purpose Abramson developed a protocol called Aloha.

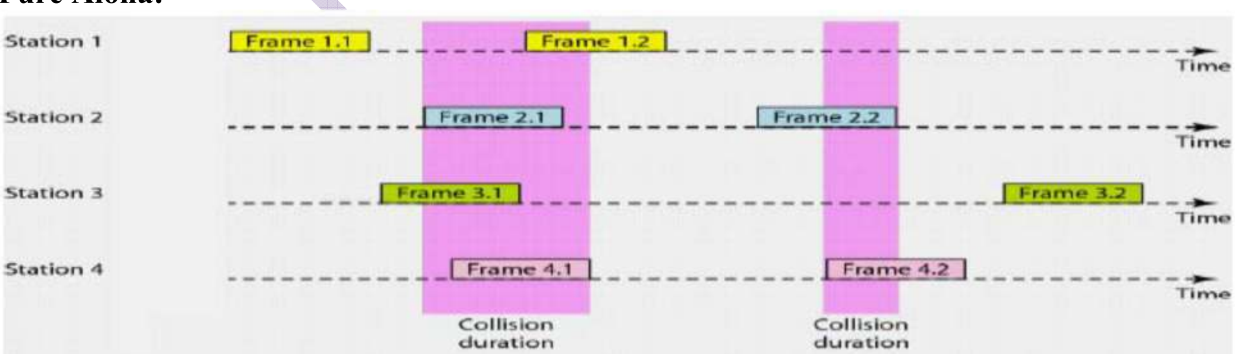
Type of Aloha
1. Pure Aloha
2. Slotted Aloha

Pure Aloha:-This simple version is also called pure ALOHA. Note that, in Pure ALOHA, sender does not check whether the channel is busy before transmitting.

- In pure ALOHA, the stations transmit frames whenever they have data to send.
- When two or more stations transmit at the same time, there will be a collision and the frames will get destroyed.
- In pure ALOHA, whenever any station transmits a frame, it expects the acknowledgement from the receiver.
- If acknowledgement is not received within specified time, the station assumes that the frame has been destroyed.
- If the frame is destroyed because of collision the station waits for a random amount of time and sends it again. This waiting time must be random otherwise same frames will collide again and again.

Therefore pure ALOHA dictates that when time-out period passes, each station must wait for a random amount of time before resending its frame. This randomness will reduce collisions.

Pure Aloha:



- The maximum throughput occurs at G = 1/2, which is 18.4%.
- In Pure Aloha, the probability of successful transmission of the data frame is S= G* e^-2G

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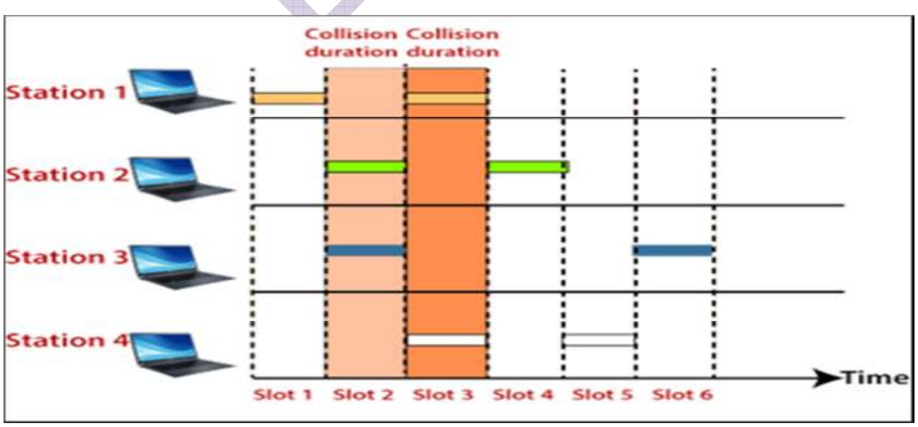
- In Pure Aloha, vulnerable time is: 2 * Tfr

Efficiency of Pure ALOHA:
Spure= G * e^-2G
where G is number of stations wants to transmit in Tt slot.
Maximum Efficiency:
Maximum Efficiency will be obtained when G=1/2
(Spure)max = 1/2 * e^-1
= 0.184
Which means, in Pure ALOHA, only about 18.4% of the time is used for successful transmissions.

Slotted Aloha:
Slotted ALOHA is an improvement over the pure ALOHA. Slotted ALOHA was invented to improve the efficiency of pure ALOHA as chances of collision in pure ALOHA are very high. In slotted ALOHA, the time of the shared channel is divided into discrete intervals called slots. The size of the slot is equal to the frame size.

In slotted ALOHA, the stations can send a frame only at the beginning of the slot and only one frame is sent in each slot. If any station is not able to place the frame onto the channel at the beginning of the slot i.e. it misses the time slot then the station has to wait until the beginning of the next time slot.



- In Slotted Aloha, maximum throughput occurs at G = 1, which is 37%.

- In Slotted Aloha, the probability of successful transmission of the data frame: S=G* e^-G
- In Slotted Aloha, vulnerable time is: Tfr

Efficiency of Slotted ALOHA:
Slotted = G * e^-G
Maximum Efficiency:
(Sslotted)max = 1 * e^-1
= 1/e = 0.368
Maximum Efficiency, in Slotted ALOHA, is 36.8%.

Ethernet 802.3 CSMA/CD:-

- IEEE 802.3 is a set of standard protocol that define Ethernet based network.
- IEEE 802.3 defined the physical layer and media access control for wired Ethernet network.
- There are a number of versions of IEEE 802.3 protocol.

The most popular ones are:

- IEEE 802.3: This standard gives for 10BASE5. It is for thick Ethernet or thick single coaxial cable.

  Note:- 10BASE5 is an early Ethernet standard that utilizes thick coaxial cable, commonly referred to as "thicknet." It supports data transfer rates of up to 10 megabits per second and has a maximum segment length of 500 meters.

- IEEE 802.3a : this gave the standard for coax cable (10BASE-2)

Note: 10Base2 is part of the IEEE 802.3 specification. It is also known as cheapernet or thinnet. The name 10Base2 is derived from several characteristics of the physical medium. The 10 comes from the maximum transmission speed of 10 Mbit/s.

- IEEE 802.3 i : this gave the standard for twisted pair (10BASE-T) that uses unshelled twisted pair(UTP)
  10BASE-T supports 10 megabits per second (Mbps) transmission speed over twisted-pair cabling with a maximum length of 100 meters (m). The twisted-pair cables connect with an RJ45 connector. 10BASE-T is a shorthand identifier designated by IEEE. The 10 refers to a maximum transmission speed of 10 Mbps.

- IEEE 802.3 j: This gave the standard for Ethernet over fiber(10BASE-F) that use fibre optics cable.

CSMA/CD:-

- Carrier sense multiple access with collision detection (CSMA/CD) is a network protocol for carrier transmission that operate in MAC address.
- It sense or listens whether the shared channel for transmission is busy or not, the collision detection technology detects collision and stop transmitting, send a jam signal and then wait for random time before transmitting.

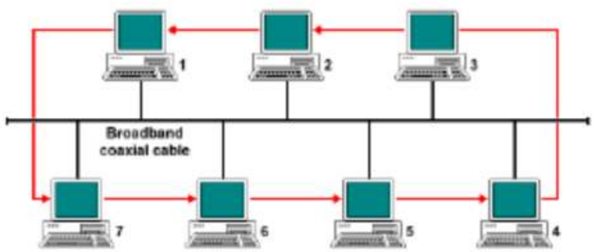
Algorithm CSMA/CD:-

- When frame is ready to transmit, the transmitting station checks whether the media is idle or busy.
- If the media is busy, the station waits until to become idle.
- If media is idle, the station start transmitting and monitors the channel to detect collision.
- If a collision is detected, the station starts the collision resolution mechanism and retransmit the frame

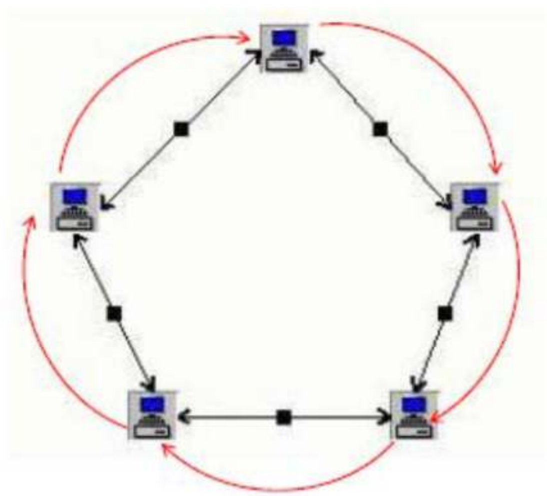
802.4 Token bus:-

- ❖ The 802.4 IEEE standards define the Token Bus protocol for a token-passing access method on a bus topology.
- ❖ In a token-passing access method, a special packet called a token is passed from station to station and only the token holder is permitted to transmit packets onto the LAN.
- ❖ No collisions can occur with this protocol (Only One Station can transfer)
- ❖ When a station is done transmitting its packets, it passes the token to the "next" station.
- ❖ The next station does not need to be physically closest to this one on the bus, just the next logical station.

❖ A station can hold the token for only a certain amount of time before it must pass it on - even if it has not completed transmitting all of its data.
❖ This assures access to all stations on the bus within a specified period of time.



802.5 Token Ring:-



❖ The 802.5 IEEE standard defines the Token Ring protocol which, like Token Bus, is another token passing access method, but for a ring topology.
❖ A ring topology consists of a series of individual point-to-point links that form a circle
❖ A token is passed from station to station in one direction around the ring, and only the station holding the token can transmit packets onto the ring.
❖ Data packets travel in only one direction around the ring.
❖ When a station receives a packet addressed to it, it copies the packet and puts it back on the ring.
❖ When the originating station receives the packet, it removes the packet.

BLUE TOOTH

- IEEE 802.15
- It is a wireless LAN technology using short-range radio links, intended to replace the cable(s) connecting portable and/or fixed electronic devices.
- It is an ad hoc network where devices can automatically find each other, establish connections, and
discover what they can do for each other.
- range 10-100 mtrs.
- features are robustness, low complexity, low power and low cost.
- uses a 2.4-GHz ISM band divided into 79 channels of 1 MHz each
- A Bluetooth device has a built-in short-range radio transmitter.
- It uses Frequency Hop Spread Spectrum (FHSS) to avoid any interference.

Applications

- Automatic synchronization between mobile and stationary devices
- Connecting mobile users to the internet using Bluetooth-enabled wire-bound connection ports
- Dynamic creation of private networks

Types of Bluetooth Wireless Technology

- Depending on the power consumption and range of the device, there are 3 Bluetooth Classes as:

1. Class 1: Max Power – 100mW ; Range – 100 m
2. Class : Max Power – 2.5mW ; Range – 10 m
3. Class : Max Power – 1mW ; Range – 1 m

Wi-Fi:

The IEEE 802.11 wireless LAN, also known as Wi-Fi, is the name of a popular wireless networking technology that uses radio waves to provide wireless high-speed Internet and network connections. Wi-Fi networks have no physical wired connection between sender and receiver, by using radio frequency (RF) technology (a frequency within the electromagnetic spectrum associated with radio wave propagation). When an RF current is supplied to an antenna, an electromagnetic field is created that then is able to propagate through space.

There are several 802.11 standards for wireless LAN technology, including 802.11b, 802.11a, and 802.11g. Table below summarizes the main characteristics of these standards. 802.11g is by far the most popular technology.

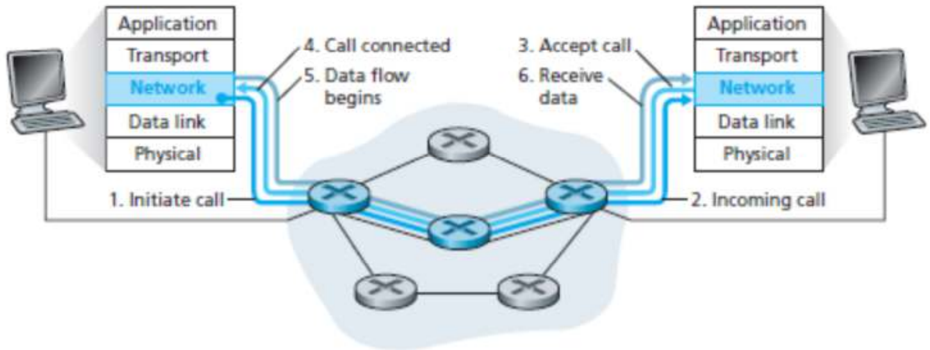
Standard	Frequency Range (United States)	Data Rate
802.11b	2.4–2.485 GHz	up to 11 Mbps
802.11a	5.1–5.8 GHz	up to 54 Mbps
802.11g	2.4–2.485 GHz	up to 54 Mbps

Wi-Fi uses multiple parts of the IEEE 802 protocol family and is designed to seamlessly interwork with its wired sister protocol Ethernet. Devices that can use Wi-Fi technologies include desktops and laptops, smartphones and tablets, smart TVs, printers, digital audio players, digital cameras, cars and drones. Compatible devices can connect to each other over Wi-Fi through a wireless access point as well as to connected Ethernet devices and may use it to access the Internet. Such an access point (or hotspot) has a range of about 20 meters (66 feet) indoors and a greater range outdoors. Hotspot coverage can be as small as a single room with walls that block radio waves, or as large as many square kilometers achieved by using multiple overlapping access points.

Wi-Fi is potentially more vulnerable to attack than wired networks because anyone within range of a network with a wireless network interface controller can attempt access. Wi-Fi Protected Access (WPA) is a family of technologies created to protect information moving across Wi-Fi networks and includes solutions for personal and enterprise networks.

Overview of Virtual Circuit Switching:

Virtual circuit switching is a packet switching methodology whereby a path is established between the source and destination through which all the packets will be routed during a call. This path is called a virtual circuit. Before the data transfer begins, the source and destination identify a suitable path for the virtual circuit. All intermediate nodes between the two points put an entry of the routing in their routing table for the call.



Overview of Frame Relay:

Frame Relay is a wide area network with the following features:

1. In 1980, Frame Relay replaced X.25

2. Frame Relay operates at a higher speed (1.544 Mbps and recently 44.376 Mbps).

3. Frame Relay operates in just the physical and data link layers. This means it can easily be used as a backbone network to provide services to protocols that already have a network layer protocol, such as the Internet.

4. Frame Relay allows bursty data.

5. Frame Relay allows a frame size of 9000 bytes, which can accommodate all local area network frame sizes.

6. Frame Relay is less expensive than other traditional WANs.

7. Frame Relay has error detection at the data link layer only. There is no flow control or error control.

Data Link Layer Protocols (DLL):

Data Link Layer (DLL) is the second layer of the OSI Model, whose main job is to send data between two devices in a reliable and error-free manner. Many types of protocols are used within this layer so that data transfer can be done in a secure, fast and organized manner.
The data link layer is divided into two sublayers:

1. **Logical Link Control (LLC) Sublayer**:

- Acts as an interface between the upper layers (network layer) and the lower layers (MAC sublayer).
- Manages flow control and error checking to ensure smooth communication.

2. Media Access Control (MAC) Sublayer:

- Handles the direct interaction with the physical layer.
- Manages how devices in a network gain access to the media and transmit data.

Functions of Logical Link Control (LLC)

- Facilitates communication between the upper layers (like the network layer) and the lower layers (like the MAC sublayer).
- Takes network protocol data and adds control information to help deliver packets to their destination.
- Implements **flow control** to manage data transmission rates and prevent data overflow.

Process:

- Receives data from the network layer and encapsulates it into frames.
- Adds headers and trailers to the frames for error detection and control.

Functions of Media Access Control (MAC) Sublayer

- Directly interacts with the physical layer through hardware, such as a **Network Interface Card (NIC)**.
- Manages **data encapsulation** (frame assembly and disassembly) and **media access control**.

Process:

- Adds a **MAC address** (a unique hardware identifier) to the frame header.
- Ensures frames are placed on the media (e.g., Ethernet cables) and removed upon reception.

- Implements **framing**, **physical addressing**, and **error control**.